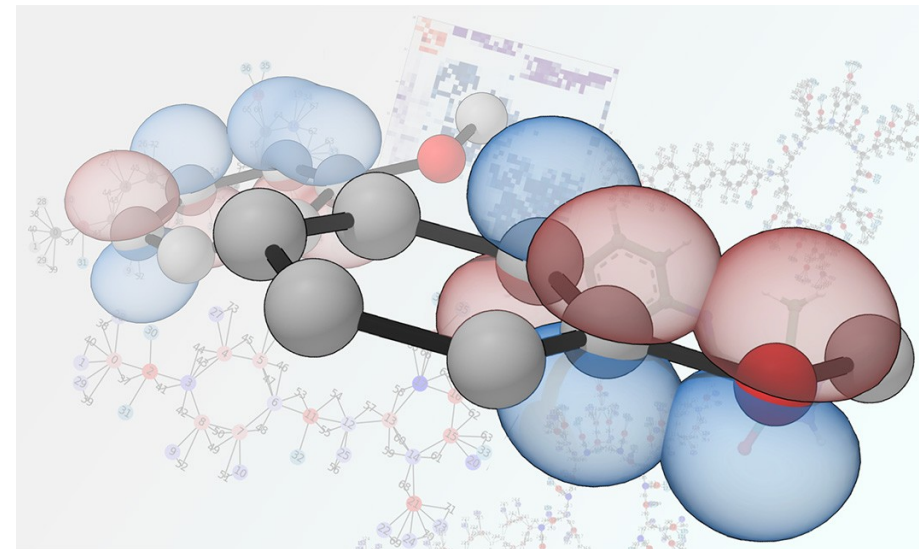


Aula 00 – Apresentação do Curso

Onde queremos chegar?



Prof. Elvis Soares
elvis@peq.coppe.ufrj.br

$$i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{H} |\Psi\rangle$$

$$|\Psi\rangle = |\psi\rangle e^{-iEt/\hbar}$$

$$\hat{H} |\psi\rangle = E |\psi\rangle$$

Infos

Período: 2026.2

Horário de aulas: 2a e 4a, 10:30-12:15

Sala de Aula: G119

Calendário: 22/Jun - 13/Set (~24 encontros)

CRITÉRIOS DE AVALIAÇÃO

- 20% de Presença e Participação
- 30% de Listas de Exercícios (~8 listas)
- 50% de Projeto Final



<https://github.com/elvissoares/COQ875-QuimicaQuantica>



<https://elvissoares.com/ensino>

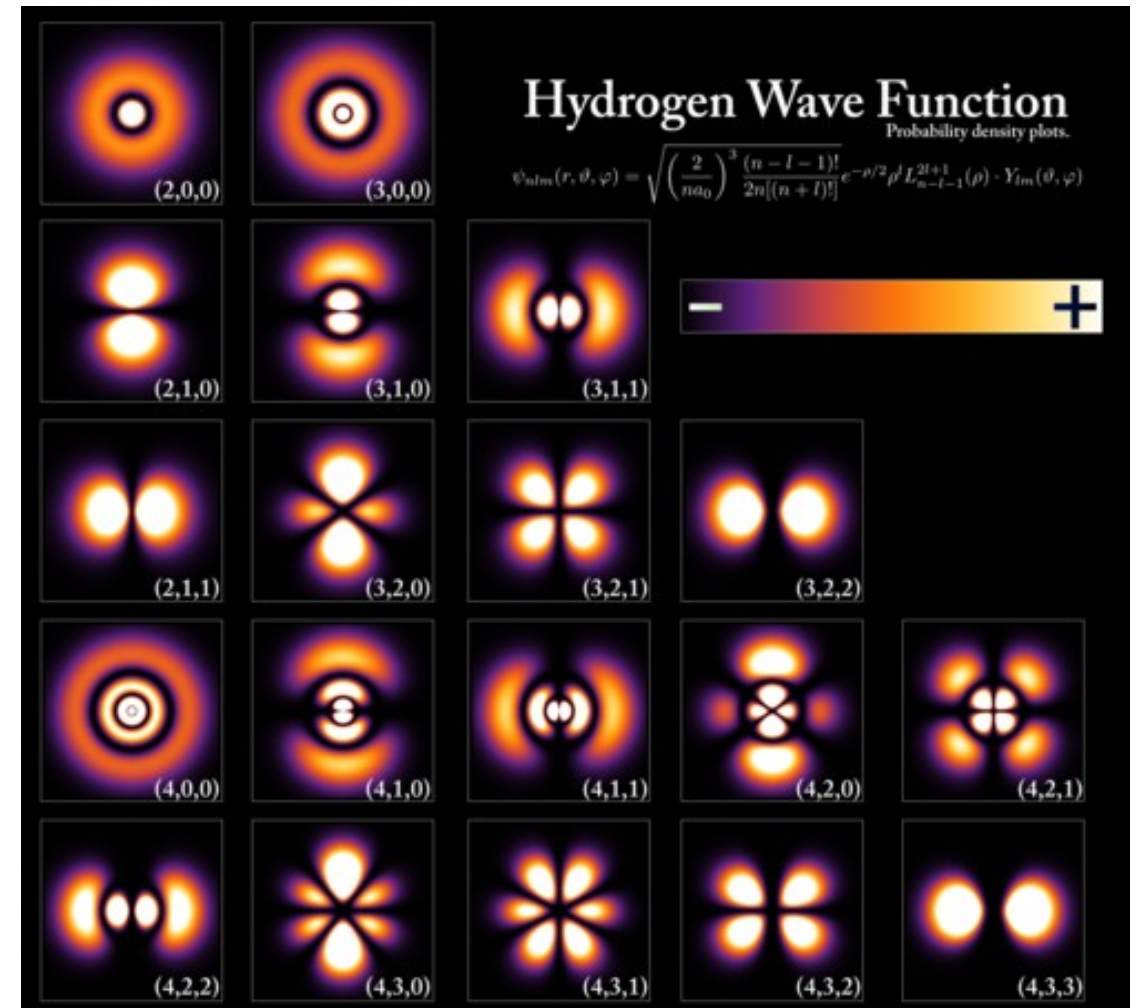
EMENTA

- 1) Fundamentos da Mecânica Quântica;
- 2) Estrutura Eletrônica de Átomos e Moléculas;
- 3) Estrutura Eletrônica de Sólidos;
- 4) Aplicações na Engenharia Química.
- 5) Caracterização de Materiais e Espectroscopia

Tópicos a serem Abordados

1) Fundamentos da Mecânica Quântica;

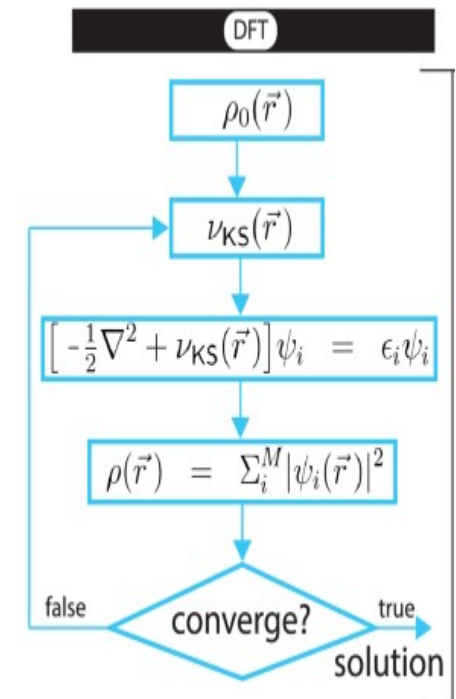
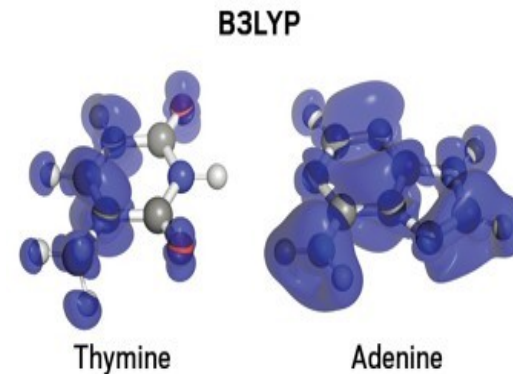
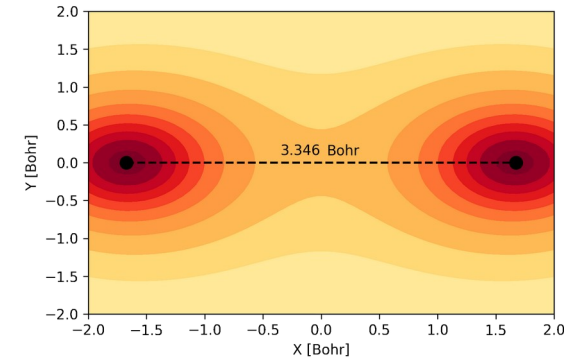
- i. Bases Experimentais e Teóricas da Mecânica Quântica
- ii. Postulados da Mecânica Quântica
- iii. Bases Matemáticas da Mecânica Quântica: Auto-valores, auto-vetores, mudança de base, comutadores
- iv. Operadores na Mecânica Quântica: posição e momento linear
- v. Equação de Schroedinger
- vi. Interpretação Probabilística da função de onda
- vii. Aplicações: Oscilador harmônico e Rotor rígido
- viii. Operador de Momento Angular
- ix. Átomo de Hidrogênio
- x. Spin



Tópicos a serem Abordados

2) Estrutura Eletrônica de Átomos e Moléculas;

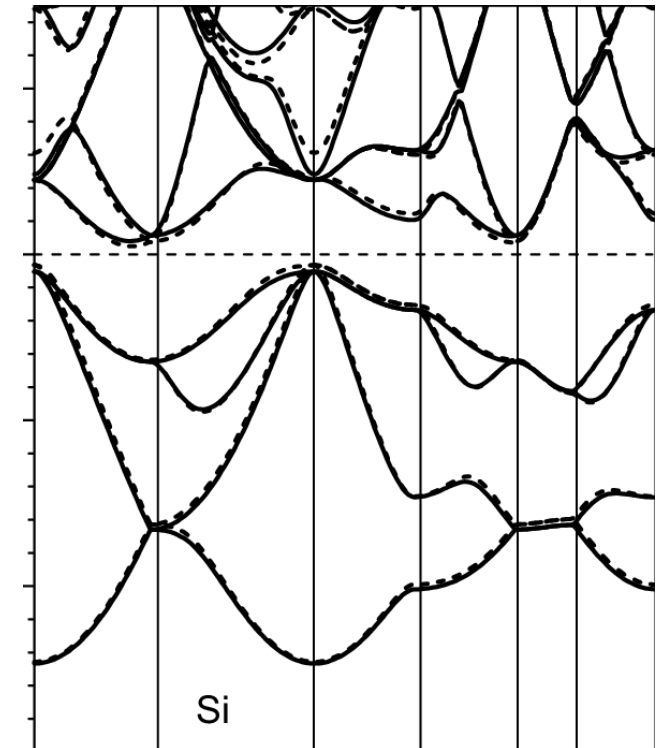
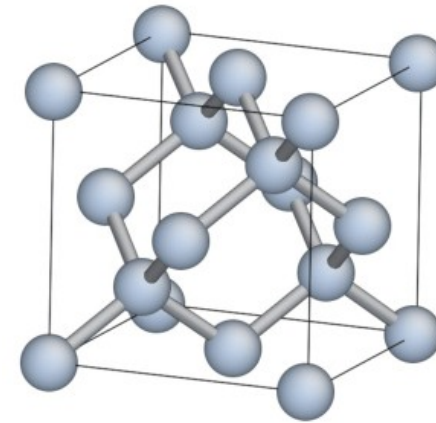
- i. Átomo de Hélio: Teorema Variacional
- ii. Molécula de H₂: Aproximação de Born-Oppenheimer e Determinante de Slater
- iii. O problema de muitos elétrons
- iv. Método de Hartree-Fock: Conjunto de Base, Equações de Roothan e método auto-consistente
- v. Discussão de métodos pós-HF
- vi. Teoria do Funcional da Densidade: Teoremas de Kohn - Hohenberg, Cálculo Variacional, Orbitais de Kohn-Sham, Funcionais de troca e correlação



Tópicos a serem Abordados

3) Estrutura Eletrônica de Sólidos;

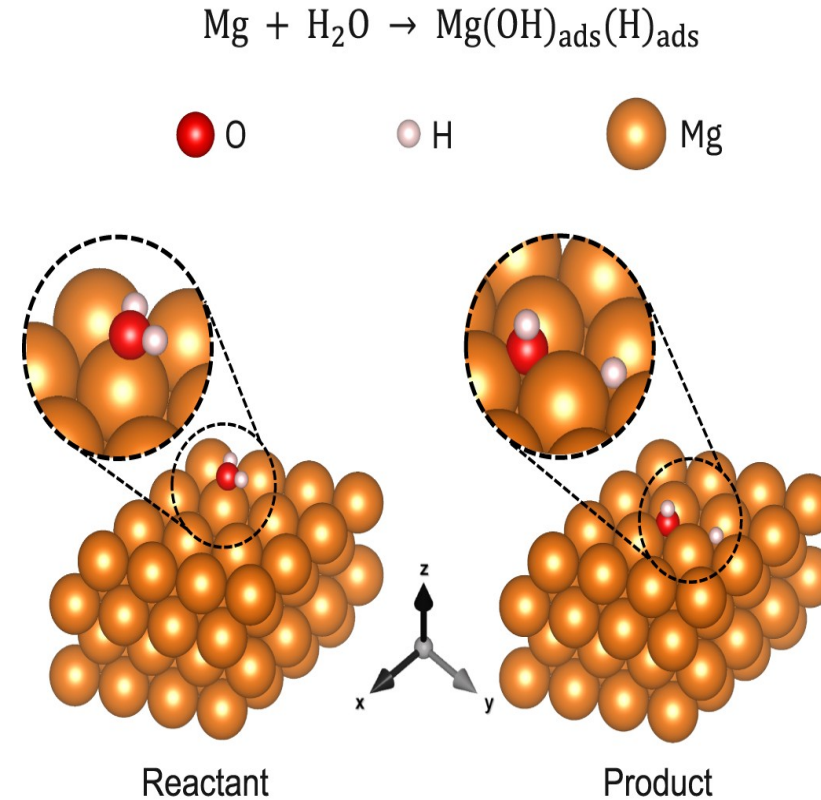
- i. Estruturas Cristalinas e Redes de Bravais
- ii. Difração de raios-X por cristais
- iii. Rede Recíproca: Transformada de Fourier, 1ª zona de Brillouin, Índices de Miller
- iv. Teorema de Bloch
- v. Densidade de Estados e Estrutura de Bandas
- vi. Projetor de onda aumentada e pseudopotenciais
- vii. Vibrações cristalinas e fônons



Tópicos a serem Abordados

4) Aplicações na Engenharia Química.

- i. Otimização de Geometria
- ii. Reação química
- iii. Adsorção
- iv. Catálise



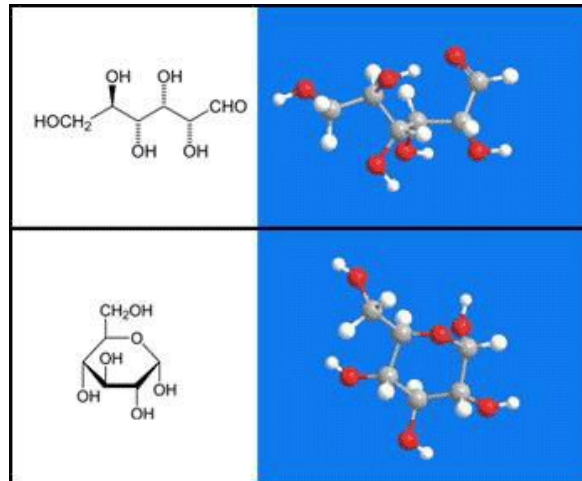
Relative electronic energy:

$$\Delta E = E_{\text{product}} - E_{\text{reactant}}$$

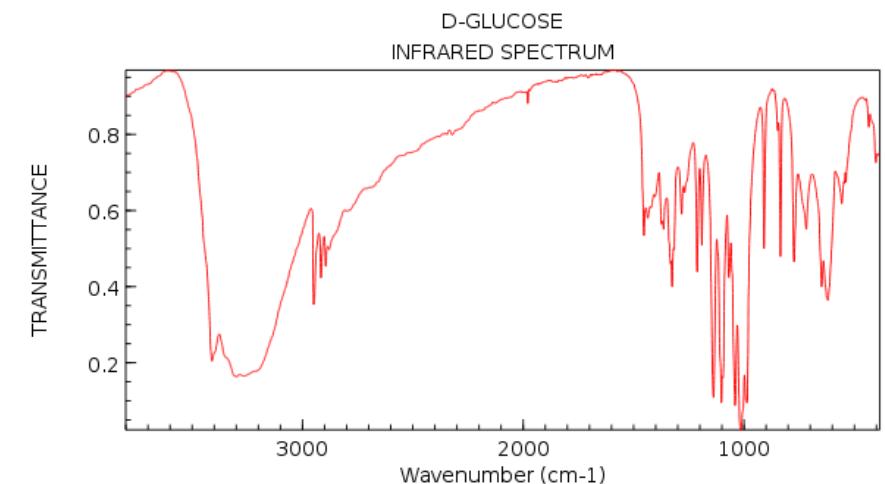
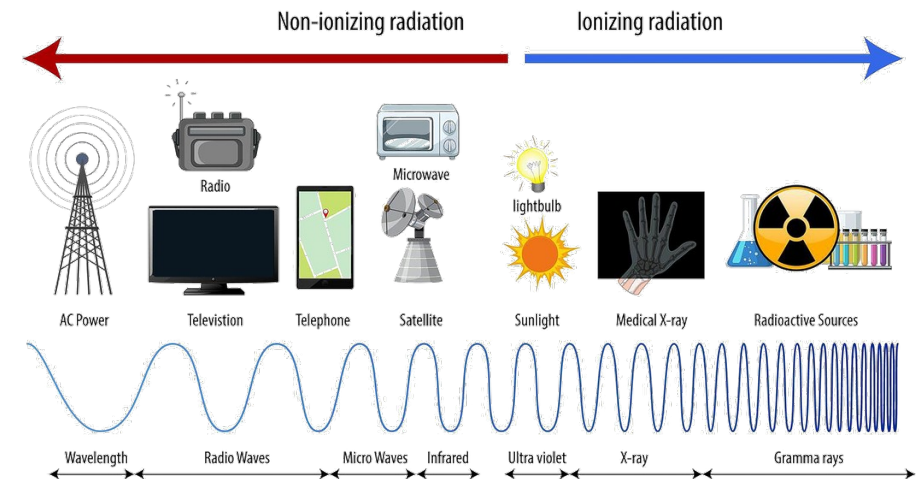
Tópicos a serem Abordados

5) Caracterização de Materiais e Espectroscopia

- i. Densidade eletrônica, momento de dipolo elétrico e polarizabilidade
- ii. Interação da radiação com a matéria
- iii. Espectro micro-ondas e a rotação
- iv. Espectro IR e Raman e a vibração
- v. Espectro UV/Vis
- vi. Calor específico de gases e sólidos
- vii. Ressonância magnética nuclear e de spin eletrônico



THE ELECTROMAGNETIC SPECTRUM

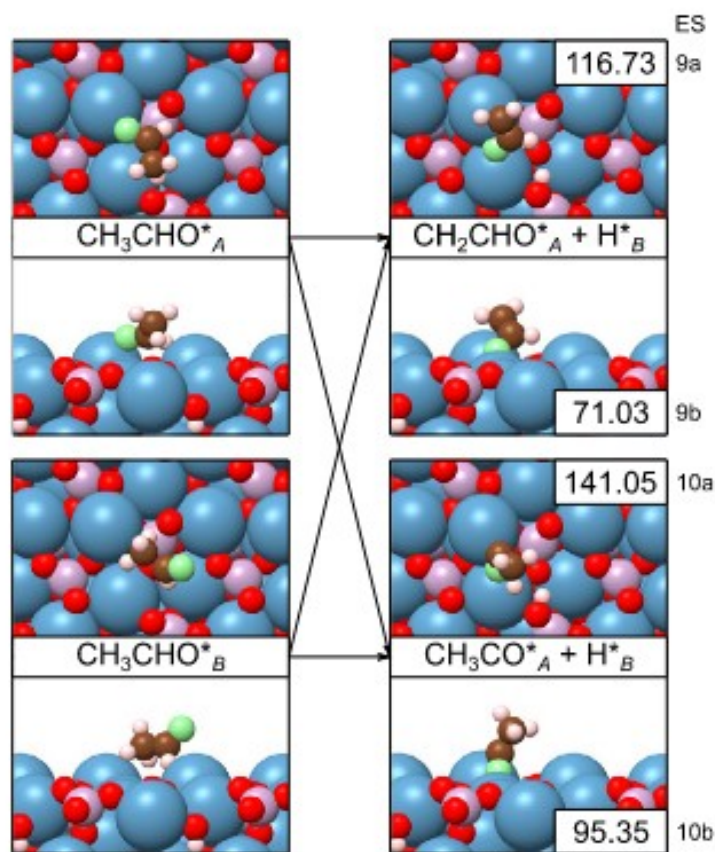


NIST Chemistry WebBook (<http://webbook.nist.gov/chemistry>)

Ferramentas Computacionais



Contexto



Brasil, H., Bittencourt, A. F. B., Yokoo, K. C. E. S., Mendes, P. C. D., Verga, L. G., Andriani, K. F., Landers, R., Da Silva, J. L. F., & Valença, G. P. (2021). Synthesis modification of hydroxyapatite surface for ethanol conversion: The role of the acidic/basic sites ratio. *Journal of Catalysis*, 404, 802–813. <https://doi.org/10.1016/j.jcat.2021.08.050>

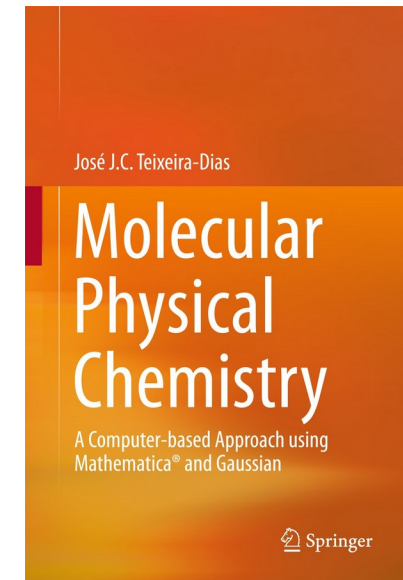
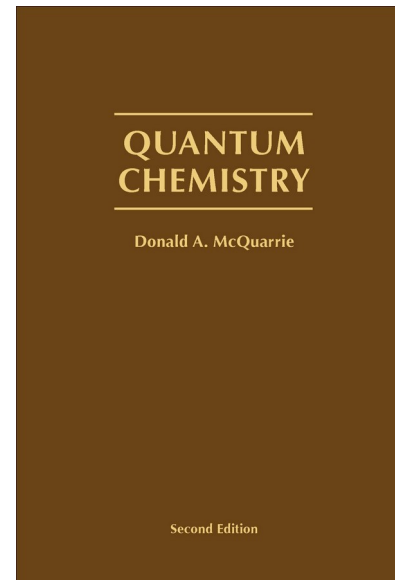
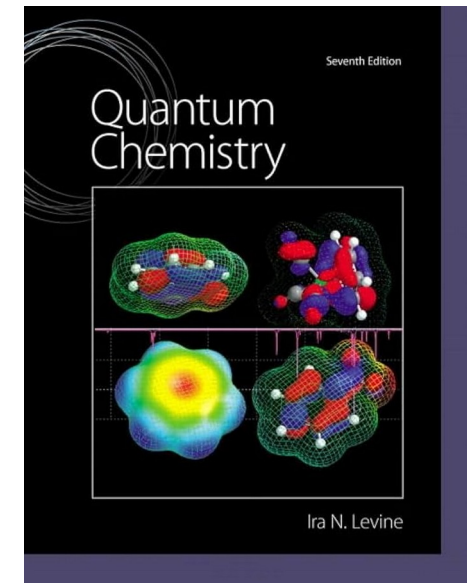
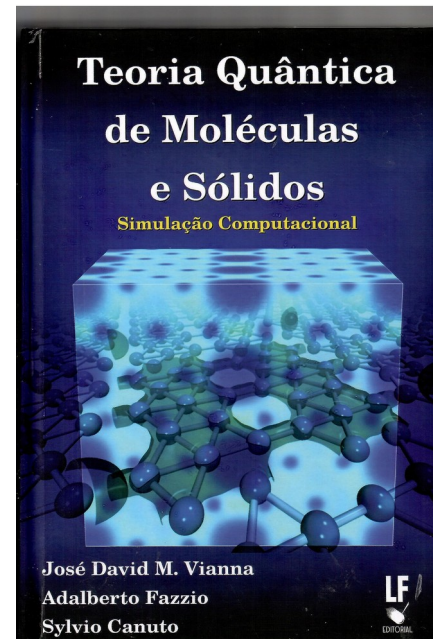
2.3. Theoretical calculations

All total energy calculations were based on spin-polarized DFT [45,46] adopting the semilocal Perdew–Burke–Ernzerhof (PBE) formulation for the exchange–correlation functional [47]. The semi-empirical D3 van der Waals (vdW) correction proposed by Grimme [48] was used to improve the description of long-range vdW interactions. We employed the Vienna *ab initio* simulation package (VASP), version 5.4.1, which implements the all-electron projected augmented-wave (PAW) method [49–52]. The electron–ion interactions were described by the PAW projectors provided within VASP along with a plane-wave cutoff energy of 489 eV, which is 12.5% higher than the largest recommended value considering all chemical elements (C, Ca, H, O, and P).

The initial geometries of the isolated gas-phase molecules were obtained from the PubChem database [53] and optimized using a 20 Å cubic box. The clean hexagonal HAP(0001) surface was modeled using a repeated slab geometry with a thickness of four layers (one formula unit, $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$, per layer and Ca/P ratio of 1.67), and a vacuum region of 15 Å. To search for the lowest energy configurations, 30 distinct structures were built for each adsorbed system using a 1×1 surface unit cell with $a_0 = b_0 = 9.497 \text{ Å}$ (88 atoms within the unit cell). The adsorption structures were built by symmetrically placing adsorbates on both sides of the slab with respect to its center of inversion. As a result, dipole correction was no longer required. Once the lowest energy configurations were identified, the optimized adsorbed structures were placed on the surface 2×2 unit cell ($a_0 = b_0 = 18.995 \text{ Å}$, 352 atoms within the unit cell) for a final structure optimization. All optimized structures were calculated allowing all atoms to relax through the conjugate gradient algorithm. The total energy convergence criterion was set to 10^{-5} eV with a force convergence of 0.025 eV Å^{-1} on each atom. Only the Γ -point was employed for the Brillouin zone integration.

Referências

- **Livro online:** <https://elvissoares.github.io/COQ875-QuimicaQuantica/>
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- McQuarrie, D. A. (2008). **Quantum chemistry, 2nd Edition**. University Science Books.
- Teixeira-Dias, J. J. (2017). **Molecular Physical Chemistry**. Springer International Publishing AG.



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- Jensen, F. (2017). **Introduction to computational chemistry**. John Wiley & Sons.
- Cramer, C. J. (2013). **Essentials of computational chemistry: theories and models**. John Wiley & Sons.
- Martin, R. M. (2020). **Electronic structure: basic theory and practical methods**. Cambridge university press.

